

ABSTRACT

The recruitment industry is increasingly driven by digital platforms, public hiring portals, and data-centric decision-making. Traditional recruitment intelligence systems require extensive manual effort to identify hiring companies, extract job requirements, monitor hiring patterns, and discover recruiter contacts. This results in inefficiencies, delayed outreach, and missed hiring opportunities.

The **AI Agentic Hiring Intelligence & Recruitment Automation System** is designed to address these challenges through an intelligent, autonomous, and scalable recruitment intelligence platform. The system integrates web scraping, artificial intelligence, analytics, and workflow automation to collect recruitment data from multiple public sources such as LinkedIn Jobs, Indeed, Naukri, startup hiring platforms, and company career portals.

The platform utilizes autonomous scraping agents to collect job-related information, including company details, job roles, required skills, recruiter contacts, salary information, and hiring locations. Artificial Intelligence models are integrated to transform unstructured job descriptions into structured recruitment intelligence by extracting roles, skills, experience requirements, and trend insights.

The system further provides hiring trend analytics, lead scoring mechanisms, recruiter intelligence modules, and role-based dashboards to support recruiters, analysts, administrators, and candidates. Secure authentication, role-based access control, protected APIs, reporting modules, and export capabilities ensure enterprise-grade operational readiness.

This project demonstrates practical implementation of modern full-stack engineering concepts including the MERN stack, JWT authentication, MongoDB database design, AI integration, browser automation, asynchronous job processing, and dashboard analytics.

The final solution offers a scalable recruitment intelligence ecosystem capable of reducing manual effort, improving hiring visibility, and enabling data-driven recruitment automation.

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INTRODUCTION

1.1 Introduction

Recruitment has evolved significantly due to digital transformation and the widespread adoption of online hiring platforms. Organizations increasingly publish job opportunities across multiple digital sources, including professional networking platforms, job boards, startup communities, and corporate career portals.

However, recruitment intelligence remains fragmented due to the following reasons:

- Job postings are distributed across multiple disconnected platforms.
- Recruiter contact information is difficult to identify manually.
- Job descriptions are largely unstructured.
- Hiring trends are not easily measurable.
- Manual company prioritization is inefficient.
- Outreach workflows are time-consuming and repetitive.

As hiring ecosystems become increasingly data-intensive, intelligent automation becomes essential.

The **AI Agentic Hiring Intelligence & Recruitment Automation System** addresses these limitations through an integrated platform that combines:

- autonomous web scraping
- structured data engineering
- AI-powered requirement extraction
- recruiter intelligence discovery
- hiring trend analytics
- lead prioritization
- recruitment workflow automation

The project demonstrates practical implementation of enterprise recruitment intelligence workflows using full-stack software engineering and applied AI.

1.2 Background

Modern recruiters rely on public hiring platforms to identify talent opportunities and business hiring trends.

Popular hiring sources include:

- LinkedIn Jobs
- Indeed
- Naukri
- Wellfound
- Startup hiring portals
- Company career pages

The challenge is not lack of data—but operational inefficiency in transforming fragmented data into actionable hiring intelligence.

Artificial Intelligence introduces opportunities to automate:

- Job requirement understanding
- Skill extraction
- Experience classification
- Role identification
- Market trend analysis
- Resume-job matching

This project leverages AI to transform raw recruitment information into structured business intelligence.

1.3 Problem Statement

Traditional recruitment intelligence processes are inefficient due to dependence on manual workflows.

Major challenges include:

Fragmented Hiring Data

Recruitment information is spread across multiple platforms, requiring repetitive manual browsing.

Unstructured Job Descriptions

Job descriptions are often written in free text, making automated analysis difficult.

Limited Recruiter Visibility

Finding recruiter or HR contact information requires significant manual effort.

Lack of Hiring Trend Intelligence

Organizations cannot easily determine:

- which companies are actively hiring
- technology demand shifts
- hiring growth trends
- geographic hiring distribution

Manual Lead Prioritization

Recruitment teams often lack objective scoring mechanisms to prioritize outreach opportunities.

Repetitive Communication Workflows

Outreach, follow-ups, and communication tracking consume operational time.

1.4 Objectives

The primary objective of this project is to develop an intelligent recruitment automation platform capable of collecting, analyzing, and operationalizing hiring intelligence.

Specific objectives include:

- To automate collection of hiring data from public sources
 - To scrape structured job and recruiter information
 - To extract structured requirements from unstructured job descriptions using AI
 - To identify recruiter and HR contact intelligence
 - To analyze hiring trends across companies
 - To rank hiring opportunities using lead scoring
 - To provide dashboard-based recruitment analytics
 - To implement secure role-based access control
 - To support exportable reporting
 - To reduce manual recruitment intelligence effort
-

1.5 Scope of the Project

The scope includes:

Included Scope

- User authentication
- role-based access control
- recruitment data scraping
- hiring data normalization
- AI requirement extraction
- recruiter intelligence discovery
- hiring trend analysis
- lead scoring
- analytics dashboards
- notifications
- reporting and export
- profile management

Optional/Extended Scope

- automated outreach
- resume-job matching
- predictive hiring intelligence
- chatbot assistant
- WhatsApp integrations
- CRM integrations

Out of Scope

- private/internal enterprise ATS integrations
 - direct payroll systems
 - applicant onboarding workflows
 - proprietary data scraping
-

1.6 Motivation

The motivation behind this project includes:

Academic Motivation

To gain practical expertise in:

- MERN stack development
- database schema design
- AI integrations
- web scraping
- API architecture
- authentication systems
- analytics dashboard development
- asynchronous processing

Industry Motivation

Recruitment technology increasingly depends on:

- automation
- AI-assisted analysis
- recruitment intelligence
- workflow optimization
-

1.7 Technologies Used

Frontend

- React.js
- Vite
- Tailwind CSS
- Redux Toolkit
- React Query
- Recharts

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose

Authentication

- JWT
- bcrypt

AI

- Google Gemini
- Groq API

Scrapping

- Playwright
- Cheerio

Background Processing

- BullMQ
- Redis

Communication

- Brevo Email Services

EXISTING SYSTEM AND PROPOSED SYSTEM

2.1 Existing System

Current recruitment intelligence processes rely heavily on manual workflows.

Typical workflow:

1. Visit job portals manually
 2. Search companies by keyword
 3. Read job descriptions manually
 4. Extract requirements manually
 5. Search recruiters separately
 6. Store data in spreadsheets
 7. Track follow-ups manually
-

2.2 Limitations of Existing System

Manual Data Collection

Repeated browsing creates inefficiency.

Time Consumption

Recruiters spend excessive time on repetitive research.

Data Inconsistency

Manual entry introduces errors.

Poor Scalability

Manual systems do not scale across hundreds of companies.

No AI Intelligence

No automated skill extraction or requirement parsing.

Weak Trend Visibility

No analytics-driven hiring intelligence.

Limited Automation

Communication workflows remain manual.

2.3 Proposed System

The proposed system introduces a unified AI-driven recruitment intelligence platform.

Capabilities include:

- autonomous source discovery
 - web scraping
 - AI extraction
 - analytics
 - lead scoring
 - dashboard visualization
 - structured reporting
 - automation workflows
-

2.4 Advantages of Proposed System

- Reduced manual effort
 - faster recruitment intelligence
 - scalable data collection
 - structured job intelligence
 - AI-powered requirement extraction
 - better lead prioritization
 - trend visibility
 - secure access control
 - centralized reporting
-

2.5 Existing vs Proposed Comparison

Parameter	Existing System	Proposed System
Data Collection	Manual	Automated
JD Analysis	Manual	AI-powered
Recruiter Discovery	Manual	Automated
Trend Analysis	Limited	Real-time
Lead Prioritization	Subjective	Scored
Reporting	Spreadsheet-based	Dashboard
Scalability	Low	High
Security	Minimal	Enterprise-grade
Automation	Low	High

REQUIREMENTS ANALYSIS

3.1 Introduction

Requirement analysis is a critical phase in software engineering that identifies the operational expectations, system capabilities, constraints, and performance requirements of the application.

The **AI Agentic Hiring Intelligence & Recruitment Automation System** is designed as a scalable intelligent recruitment intelligence platform. The requirement analysis defines both the functional and non-functional capabilities required to achieve the intended business objectives.

3.2 Functional Requirements

Functional requirements describe the specific services and operations that the system must perform.

3.2.1 User Authentication Module

The system shall provide:

- User registration
- Secure login functionality
- Email verification using OTP
- JWT-based authentication
- Refresh token support
- Secure logout functionality
- Password hashing using bcrypt
- Role-based access authorization
- Protected API access

Supported roles:

- Administrator
 - Recruiter
 - Analyst
 - Candidate/User
-

3.2.2 Profile Management Module

The system shall allow users to:

- Create profile information
 - Update personal details
 - Upload resumes
 - Manage skill preferences
 - Configure dashboard preferences
 - Manage notification settings
-

3.2.3 Hiring Data Collection Module

The platform shall support automated hiring data collection from public sources.

Functions:

- Search hiring sources
- Extract job listings
- Collect company information
- Capture recruiter information
- Extract salary details
- Collect location data
- Parse posting dates
- Deduplicate repeated jobs

Supported sources:

- LinkedIn Jobs
 - Indeed
 - Naukri
 - Startup hiring portals
 - Company career websites
-

3.2.4 AI Requirement Extraction Module

The system shall:

- Accept raw job descriptions
- Analyze unstructured text
- Extract:
 - Job role
 - Required skills
 - Experience range
 - Salary indicators
 - Technology stack
 - Hiring priorities

AI providers:

- Gemini
 - Groq
-

3.2.5 Recruiter Intelligence Module

The platform shall:

- Discover recruiter contacts
- Store recruiter information
- Associate recruiters with companies
- Maintain recruiter directories
- Support recruiter intelligence lookups

Captured information:

- Name
 - Designation
 - Email
 - LinkedIn
 - Source
-

3.2.6 Hiring Trend Analysis Module

The system shall:

- Analyze hiring frequency
- Detect active hiring organizations
- Track technology demand
- Detect expansion signals
- Monitor hiring trends over time

Outputs:

- trend classification
 - trend score
 - technology demand metrics
-

3.2.7 Lead Scoring Module

The system shall prioritize companies based on measurable parameters.

Scoring factors:

- Active openings
 - Hiring frequency
 - Skill relevance
 - Recruiter availability
 - Growth indicators
 - Trend intelligence
-

3.2.8 Reporting Module

The system shall generate reports for analytics and operational use.

Supported exports:

- CSV
- Excel
- PDF

Reports:

- Company analytics
 - Skill demand
 - Hiring trends
 - Recruiter intelligence
 - Lead scoring reports
-

3.2.9 Dashboard Module

Dashboards shall provide:

- hiring statistics
 - analytics charts
 - recruiter metrics
 - recent activity
 - trend visualization
 - role-specific actions
-

3.2.10 Notification Module

The system shall support:

- real-time alerts
 - scraping completion notifications
 - AI extraction updates
 - lead score notifications
 - account activity alerts
-

3.2.11 Outreach Automation Module (Extended Scope)

Optional automation support:

- email outreach
 - follow-up scheduling
 - communication logging
-

3.3 Non-Functional Requirements

Non-functional requirements define operational quality attributes.

3.3.1 Performance

System shall:

- handle concurrent users efficiently
- process scraping tasks asynchronously
- provide responsive dashboards
- minimize API latency

Target expectations:

- API response < 3 seconds
 - dashboard loading < 5 seconds
 - background scraping independently processed
-

3.3.2 Scalability

The platform shall support:

- increasing user volume
- increasing scraping workloads
- larger hiring datasets
- future module expansion

Scalability strategies:

- modular architecture
 - service separation
 - Redis queues
 - background workers
-

3.3.3 Security

Security requirements:

- encrypted passwords

- JWT validation
 - secure authentication
 - RBAC authorization
 - input validation
 - rate limiting
 - secure headers
 - CORS protection
-

3.3.4 Reliability

System shall ensure:

- stable API operations
 - background job resilience
 - retry handling
 - duplicate prevention
 - graceful failure recovery
-

3.3.5 Maintainability

System design must support:

- modular code organization
 - reusable services
 - clean architecture
 - extensibility
-

3.3.6 Availability

Expected operational behavior:

- reliable API uptime
 - accessible dashboards
 - resilient background processes
-

3.3.7 Usability

Frontend shall provide:

- intuitive dashboards
 - responsive layouts
 - role-specific navigation
 - clear visual analytics
-

3.4 Software Requirements

Frontend

- React.js
 - Vite
 - Tailwind CSS
 - Redux Toolkit
 - React Query
 - Framer Motion
 - Recharts
-

Backend

- Node.js
 - Express.js
-

Database

- MongoDB
 - Mongoose
-

Authentication

- JWT
- bcryptjs

AI Services

- Gemini API
 - Groq API
-

Scrapping Tools

- Playwright
 - Cheerio
-

Background Processing

- BullMQ
 - Redis
-

Email Services

- Brevo
-

3.5 Hardware Requirements

Minimum development environment:

Processor

Intel Core i5 / equivalent

RAM

8 GB minimum

Storage

20 GB free

Internet

Stable broadband connection

Browser

Chrome / Edge / Firefox

3.6 Use Case Analysis

Actors

Primary actors:

- Administrator
 - Recruiter
 - Analyst
 - Candidate/User
 - AI Engine
 - Scraping Agent
-

3.7 Use Cases

Administrator

Actions:

- manage users
 - monitor scraping
 - view analytics
 - manage system settings
 - access reports
 - manage jobs
-

Recruiter

Actions:

- search jobs
 - trigger AI extraction
 - view leads
 - access recruiter intelligence
 - initiate outreach
 - generate reports
-

Analyst

Actions:

- analyze hiring trends
 - view dashboards
 - generate analytics
 - study technology demand
-

Candidate/User

Actions:

- manage profile
 - upload resume
 - search opportunities
 - view recommendations
-

AI Engine

Responsibilities:

- parse job descriptions
 - extract structured requirements
 - analyze intelligence
 - support scoring
-

Scraping Agent

Responsibilities:

- discover sources
- collect hiring data
- normalize content
- store structured records

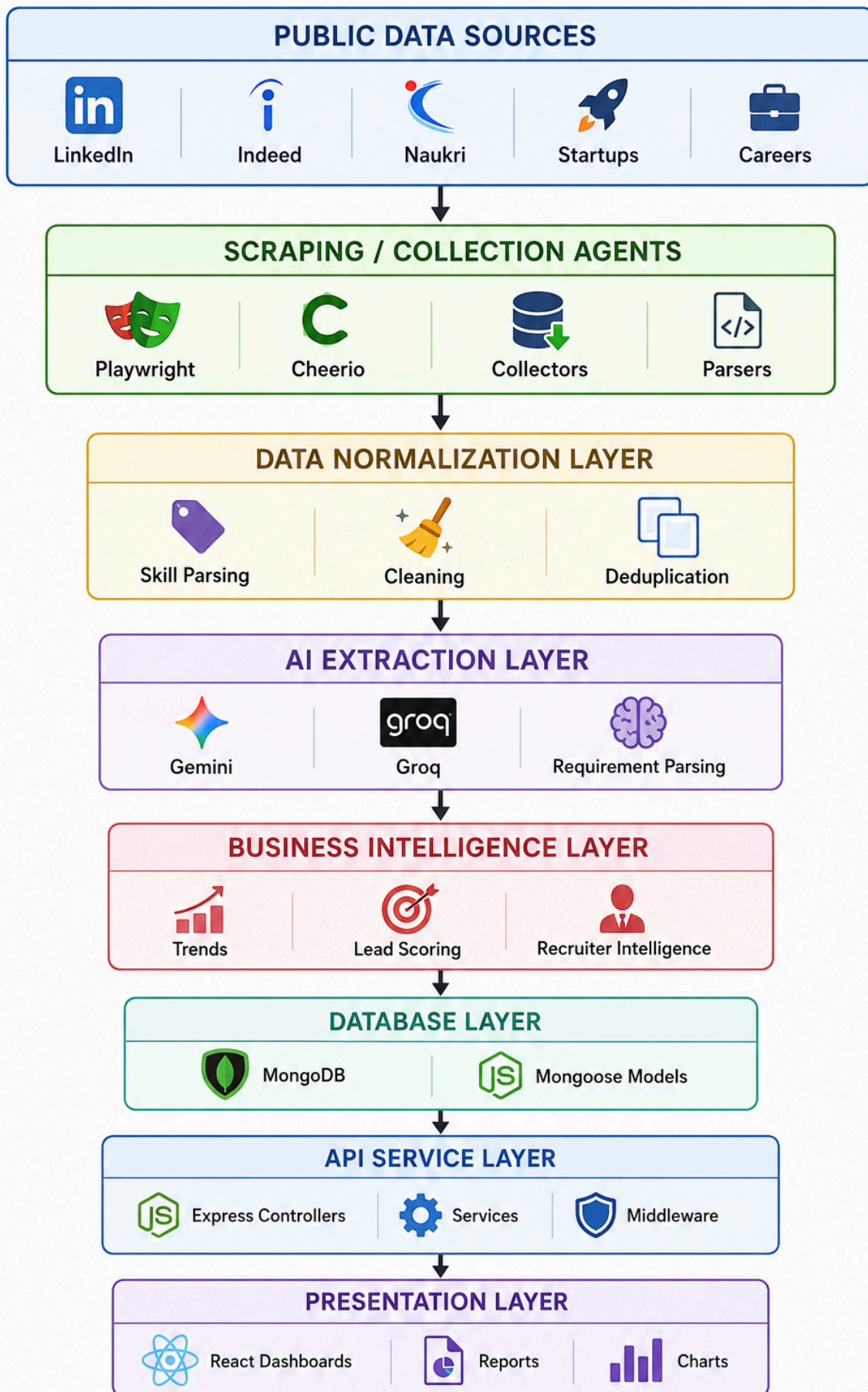
SYSTEM ARCHITECTURE & DESIGN

4.1 Introduction

The architecture follows a modular service-oriented design integrating:

- frontend presentation layer
- backend API layer
- database layer
- AI intelligence layer
- scraping agents
- asynchronous processing
- analytics services

4.2 High-Level Architecture



4.3 Architectural Components

Frontend Layer

Responsibilities:

- user interaction
- dashboards
- analytics visualization
- forms
- protected navigation
- API communication

Technologies:

- React
 - Redux Toolkit
 - Tailwind CSS
 - Recharts
 - React Query
-

Backend Layer

Responsibilities:

- request handling
- authentication
- business logic
- API orchestration
- validation
- security enforcement

Technologies:

- Node.js
 - Express.js
-

Database Layer

Responsibilities:

- persistent storage
- structured collections
- relationship management
- analytics storage

Technology:

- MongoDB
-

AI Layer

Responsibilities:

- JD parsing
- skill extraction
- structured requirement conversion
- AI intelligence workflows

Technologies:

- Gemini
 - Groq
-

Scrapping Layer

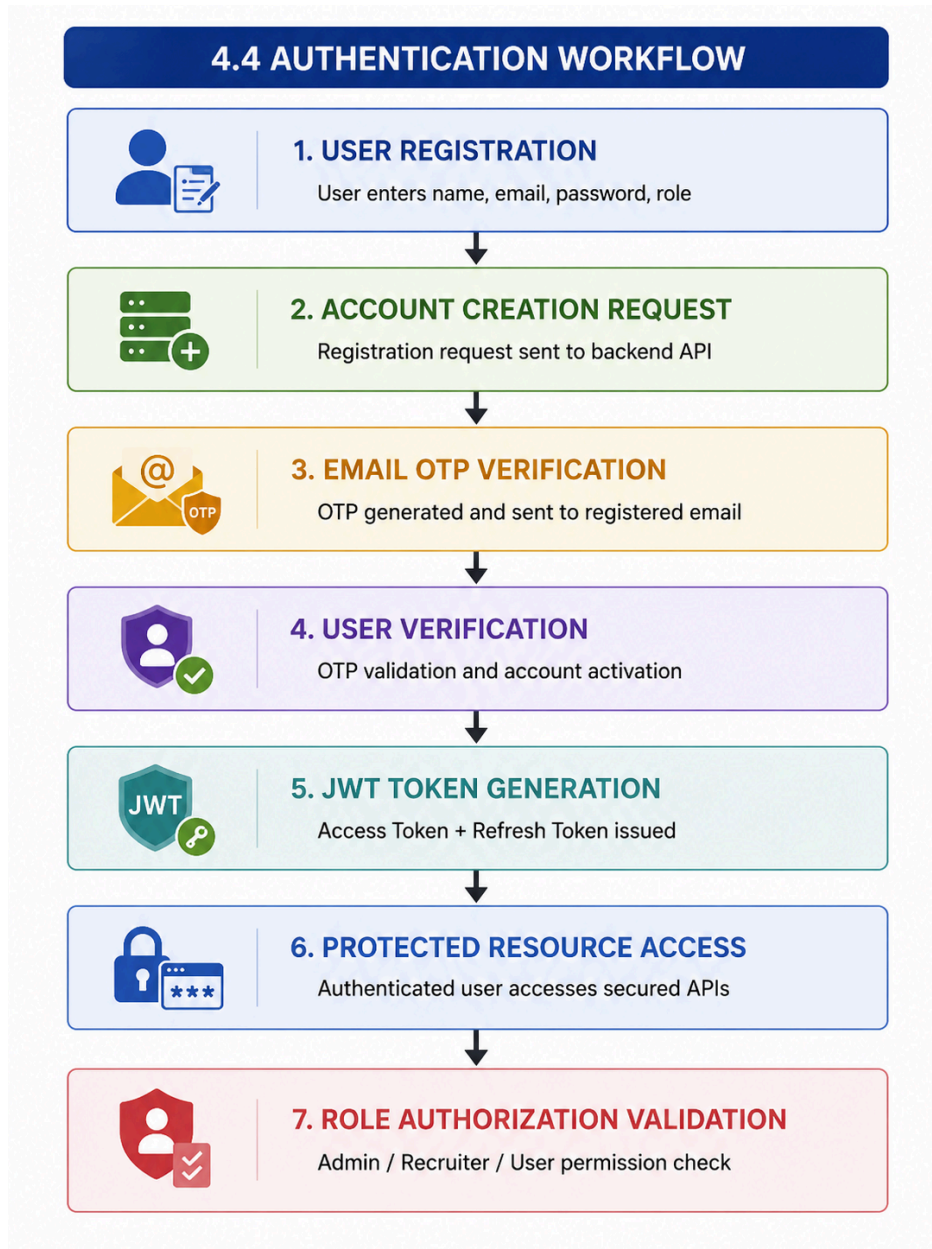
Responsibilities:

- public data collection
- DOM parsing
- automation workflows
- content extraction

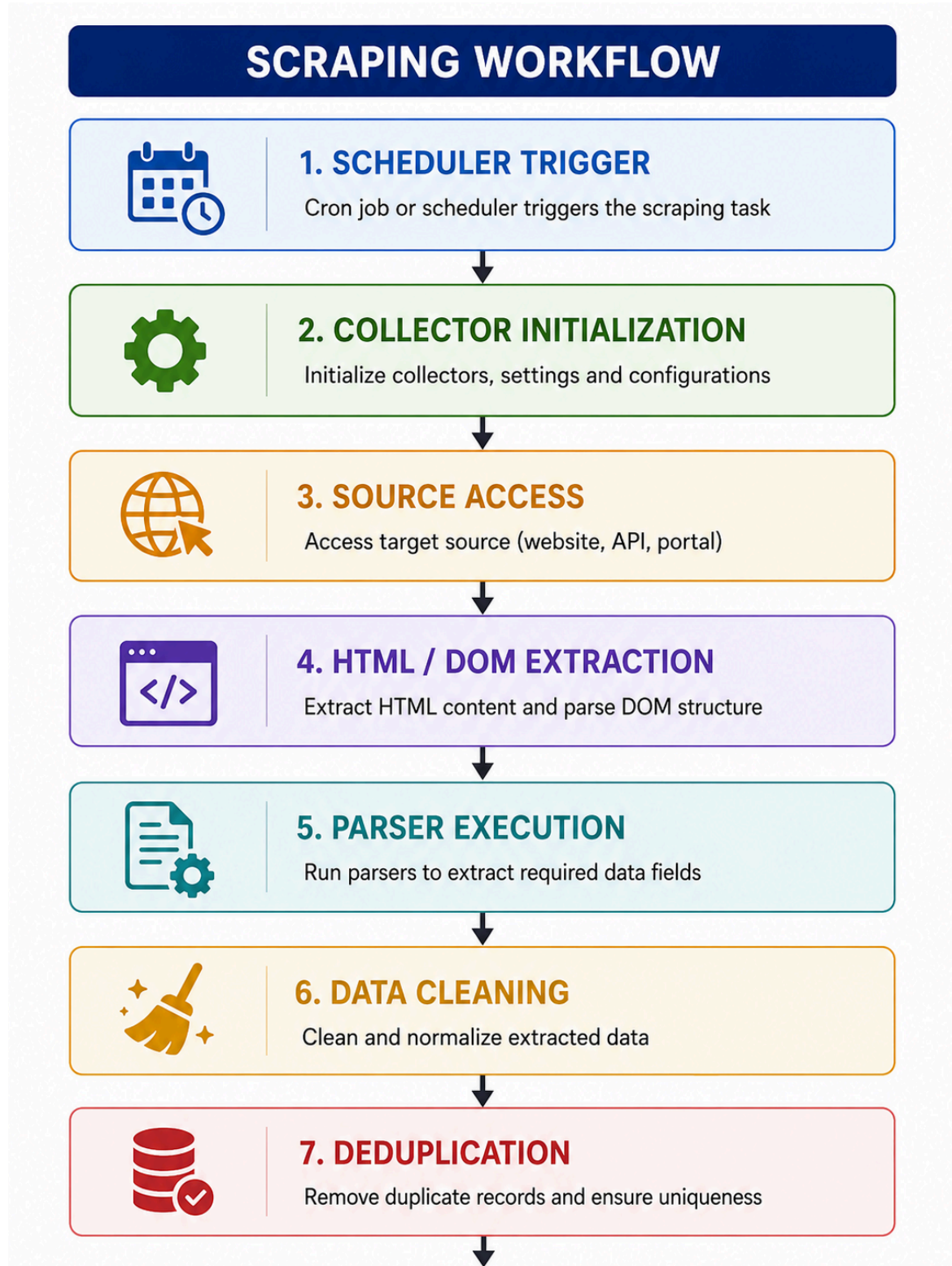
Technologies:

- Playwright
- Cheerio

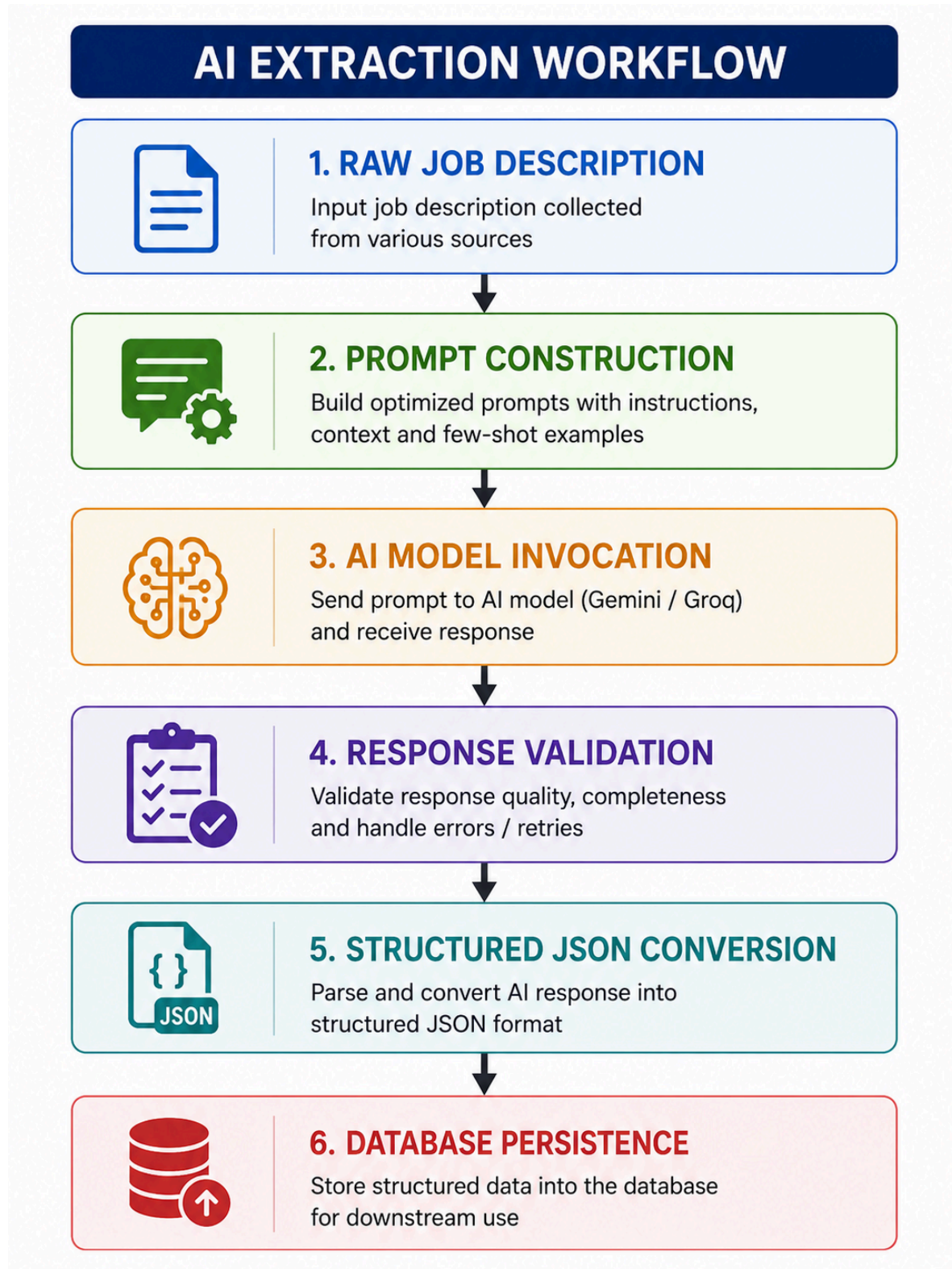
4.4 Authentication Workflow



4.5 Scrapping Workflow



4.6 AI Extraction Workflow



4.7 Lead Scoring Workflow



4.8 Dashboard Workflow



DATABASE DESIGN

5.1 Introduction

Database design plays a foundational role in ensuring scalability, maintainability, and data consistency for the recruitment intelligence platform.

The system uses **MongoDB**, a NoSQL document-oriented database, due to its flexibility in handling semi-structured and evolving recruitment datasets such as job descriptions, recruiter intelligence, analytics outputs, and AI-generated structured responses.

Database interaction is managed through **Mongoose ODM**, enabling schema definition, validation, indexing, and relationship mapping.

5.2 Database Technology Justification

Why MongoDB?

MongoDB was selected for the following reasons:

- Flexible schema support
 - Efficient handling of semi-structured scraped data
 - JSON-like document storage
 - High scalability
 - Rapid development support
 - Native integration with Node.js ecosystem
 - Suitable for analytics and evolving AI outputs
-

Why Mongoose?

Advantages:

- schema enforcement
- validation support
- middleware support
- relationship references
- timestamps
- indexing
- query abstraction

5.3 Core Database Collections

The application consists of multiple collections supporting user management, recruitment intelligence, analytics, AI extraction, and operational workflows.

5.4 Users Collection

Purpose

Stores system user authentication and authorization information.

Fields

Field Name	Type	Description
_id	ObjectId	Unique identifier
fullName	String	User full name
email	String	Unique email
password	String	Hashed password
role	String	admin / recruiter / analyst / user
isVerified	Boolean	Email verification status
refreshToken	String	JWT refresh token
createdAt	Date	Record creation
updatedAt	Date	Record update

Sample Document

```
{
  "_id": "665001",
  "fullName": "Harsha",
  "email": "harsha@example.com",
  "password": "hashed_password",
```

```
"role": "recruiter",
"isVerified": true
}
```

5.5 Company Collection

Purpose

Stores hiring company intelligence.

Fields

Field	Type	Description
_id	ObjectId	Primary key
companyName	String	Company name
website	String	Official website
industry	String	Domain category
hiringScore	Number	Lead priority score
trend	String	Hiring trend label
createdAt	Date	Created timestamp

Sample Document

```
{
  "companyName": "TechNova Solutions",
  "website": "https://technova.com",
  "industry": "Software",
  "hiringScore": 91,
  "trend": "Aggressive Hiring"
}
```

5.6 HiringJob Collection

Purpose

Stores raw scraped job information.

Fields

Field	Type
companyId	ObjectId
source	String
jobRole	String
rawDescription	String
jobUrl	String
location	String
salary	String
postedDate	Date
hash	String

Purpose of Hash Field

Used for duplicate detection across sources.

5.7 Job Collection

Purpose

Stores normalized structured jobs for analytics.

Fields

Field	Type
companyId	ObjectId
role	String
skills	Array
experience	String
location	String
salary	String
postedDate	Date

5.8 Recruiter Collection

Purpose

Stores recruiter intelligence.

Fields

Field	Type
companyId	ObjectId
name	String
designation	String
email	String
linkedin	String
source	String

5.9 RequirementExtraction Collection

Purpose

Stores AI-structured extraction outputs.

Fields

Field	Type
userId	ObjectId
jobId	ObjectId
extractedRole	String
extractedSkills	Array
extractedExperience	String
salaryInsights	String
rawAIResponse	Object

5.10 LeadScore Collection

Purpose

Stores company prioritization intelligence.

Fields

Field	Type
companyId	ObjectId
score	Number
hiringFrequencyScore	Number
technologyMatchScore	Number
recruiterAvailabilityScore	Number
growthSignalScore	Number

5.11 Trend Collection

Purpose

Stores hiring trend intelligence.

Fields

Field	Type
companyId	ObjectId
trendType	String
score	Number
analysis	String

5.12 CandidateProfile Collection

Purpose

Stores candidate professional information.

Fields

Field	Type
userId	ObjectId
skills	Array
expectedSalary	String
preferredRoles	Array
experience	String
resumeUrl	String

5.13 RecruiterProfile Collection

Purpose

Stores recruiter-specific metadata.

Fields

Field	Type
userId	ObjectId
companyName	String
designation	String
contactNumber	String

5.14 AnalystProfile Collection

Purpose

Stores analyst specialization.

Fields

Field	Type
userId	ObjectId
industriesCovered	Array
specialization	String
yearsExperience	String

5.15 Notification Collection

Purpose

Stores user notification events.

Fields

Field	Type
userId	ObjectId
title	String
message	String
read	Boolean
createdAt	Date

5.16 OutreachLog Collection

Purpose

Tracks outreach communication.

Fields

Field	Type
recruiterId	ObjectId
companyId	ObjectId
channel	String
status	String
sentAt	Date

5.17 Resume Collection

Purpose

Stores uploaded resumes.

Fields

Field	Type
userId	ObjectId
fileUrl	String
parsedSkills	Array
uploadDate	Date

5.18 SavedSearch Collection

Purpose

Stores user search preferences.

Fields

Field	Type
userId	ObjectId
filters	Object
createdAt	Date

5.19 Relationship Model

One-to-One

- User ↔ CandidateProfile
 - User ↔ RecruiterProfile
 - User ↔ AnalystProfile
 - User ↔ Resume
-

One-to-Many

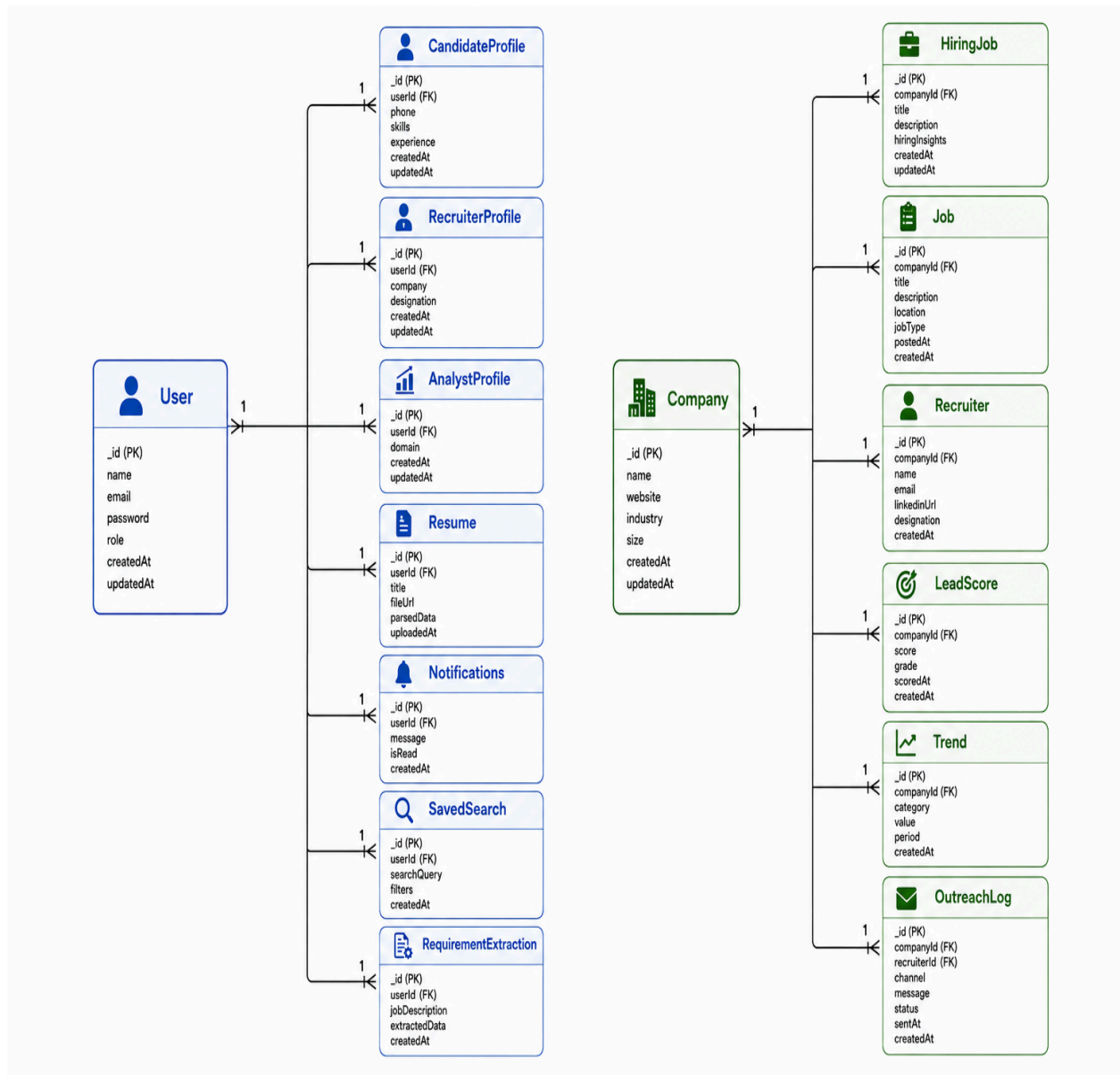
- Company → Jobs
 - Company → Recruiters
 - Company → LeadScores
 - Company → Trends
 - User → Notifications
 - User → SavedSearch
 - User → AI Extractions
-

Many-to-Many (Logical)

- Jobs ↔ Skills

Implemented via arrays.

5.20 ER Diagram



MODULE IMPLEMENTATION

6.1 Authentication Module

Purpose

Provides secure user access control.

Features

- registration
- OTP verification
- login
- logout
- refresh token
- RBAC
- password encryption

Workflow

1. User registers
 2. OTP sent via email
 3. Verification performed
 4. JWT generated
 5. Protected access granted
-

6.2 Role-Based Access Control Module

Roles

- Admin
- Recruiter
- Analyst
- User

Authorization Strategy

Middleware validates:

- token authenticity

- user existence
 - role permissions
-

6.3 Hiring Data Collection Module

Purpose

Collect public hiring intelligence.

Tools

- Playwright
- Cheerio

Workflow

1. collector initialization
 2. source browsing
 3. DOM extraction
 4. parser processing
 5. normalization
 6. duplicate filtering
 7. storage
-

6.4 AI Requirement Extraction Module

Purpose

Convert raw JD text into structured intelligence.

AI Inputs

Raw job description text.

AI Outputs

- role
- skills
- experience

- salary indicators
- tech stack

Providers

- Gemini
 - Groq
-

6.5 Recruiter Intelligence Module

Purpose

Identify recruiter contacts.

Outputs

- recruiter name
 - email
 - linkedin
 - source attribution
-

6.6 Hiring Trend Analysis Module

Purpose

Analyze company hiring behavior.

Logic

Inputs:

- frequency
- location expansion
- technology growth
- role recurrence

Outputs:

- aggressive hiring
- moderate hiring

- hiring freeze
-

6.7 Lead Scoring Module

Weighted Parameters

Parameter	Weight
Active openings	25%
Hiring frequency	20%
Technology relevance	20%
Recruiter intelligence	15%
Growth indicators	20%

6.8 Dashboard Analytics Module

Features

- statistics cards
 - hiring trends
 - recruiter metrics
 - lead intelligence
 - reports
-

6.9 Notification Module

Purpose

System communication alerts.

Supports:

- scraping alerts
- AI processing alerts

6.10 Reporting Module

Formats

- CSV
- Excel
- PDF

Reports

- hiring analytics
 - recruiter reports
 - company scoring
 - trends
-

6.11 Resume Analysis Module

Purpose

Candidate intelligence enhancement.

Features:

- resume upload
 - parsing
 - skill extraction
 - AI scrutiny
 - recommendation logic
-

6.12 Outreach Module

Extended Functionality

Supports:

- email communication
- tracking
- logging
- follow-up automation

API DOCUMENTATION

7.1 Introduction

The backend exposes RESTful APIs to support frontend dashboards, authentication, AI workflows, scraping orchestration, analytics, and operational modules.

The APIs are secured using JWT authentication and role-based access middleware.

7.2 Authentication APIs

Method	Endpoint	Description	Access
POST	/api/auth/register	Register user	Public
POST	/api/auth/verify-email	OTP verification	Public
POST	/api/auth/login	Authenticate user	Public
POST	/api/auth/refresh	Refresh token	Authenticated
POST	/api/auth/logout	Logout user	Authenticated
GET	/api/auth/me	Fetch profile	Authenticated

Example Register Request

```
{
  "fullName": "Harsha",
  "email": "harsha@example.com",
  "password": "StrongPassword123"
}
```

Example Login Response

```
{
  "success": true,
  "token": "jwt_access_token",
  "role": "recruiter",
  "user": {
```

```
"id": "123",  
"email": "harsha@example.com"  
}  
}
```

7.3 Hiring Intelligence APIs

Method	Endpoint	Description	Access
POST	/api/hiring/collect	Trigger scraping	Admin / Recruiter
GET	/api/hiring/jobs	Fetch jobs	Authenticated
POST	/api/hiring/jobs	Manual job entry	Recruiter
GET	/api/hiring/jobs/:id	Job details	Authenticated

7.4 AI Requirement APIs

Method	Endpoint	Description	Access
POST	/api/requirement/extract	Single JD extraction	Recruiter / Analyst
POST	/api/requirement/extract-batch	Bulk extraction	Recruiter / Analyst

7.5 Recruiter Intelligence APIs

Method	Endpoint	Description	Access
GET	/api/recruiters	Recruiter directory	Recruiter / Analyst
GET	/api/recruiters/:id	Recruiter details	Recruiter / Analyst

7.6 Lead Scoring APIs

Method	Endpoint	Description	Access
POST	/api/lead-score/calculate	Generate lead score	Recruiter
GET	/api/lead-score/company/:id	View company score	Recruiter

7.7 Analytics APIs

Method	Endpoint	Description	Access
GET	/api/analytics/hiring-trends	Trend analysis	Analyst
GET	/api/analytics/tech-demand	Skill analytics	Analyst
GET	/api/analytics/company-ranking	Lead rankings	Analyst

7.8 Resume APIs

Method	Endpoint	Description	Access
POST	/api/resume/upload	Upload resume	User
POST	/api/resume/analyze	AI analysis	User

7.9 Notification APIs

Method	Endpoint	Description	Access
GET	/api/notifications	Fetch notifications	Authenticated
PATCH	/api/notifications/:id/read	Mark as read	Authenticated

FRONTEND DESIGN

8.1 Introduction

The frontend is developed using **React.js** with responsive dashboard-driven design principles.

Goals:

- intuitive navigation
 - fast interactions
 - protected access
 - role-specific workspaces
 - visual analytics
-

8.2 Public Pages

Home Page

Features:

- project overview
 - CTA buttons
 - feature highlights
-

Login Page

Features:

- email login
 - password authentication
 - validation
-

Registration Page

Features:

- account creation

- OTP verification initiation
-

Forgot Password Page

Features:

- reset initiation
-

8.3 Role-Based Dashboards

Admin Dashboard

Features:

- user management
 - scraping monitoring
 - analytics overview
 - system insights
 - operational alerts
-

Recruiter Dashboard

Features:

- hiring jobs
 - recruiter intelligence
 - lead scoring
 - AI extraction
 - outreach access
 - reports
-

Analyst Dashboard

Features:

- trend intelligence
 - technology demand
 - company analytics
 - report generation
-

Candidate Dashboard

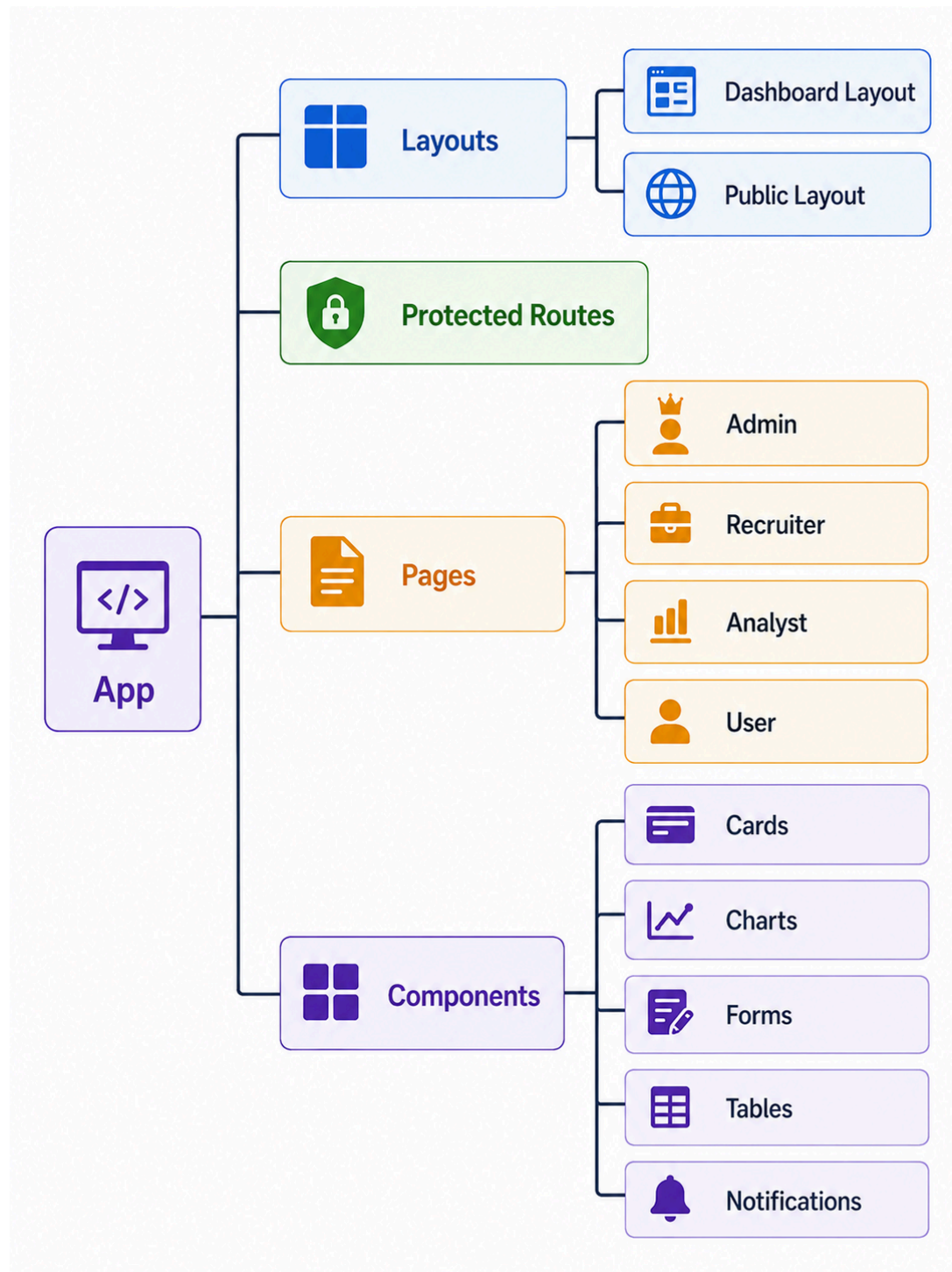
Features:

- profile management
 - resume analysis
 - job discovery
 - recommendations
-

8.4 Shared Pages

- profile setup
 - settings
 - notifications
 - reports
 - saved searches
-

8.5 Frontend Component Architecture



SECURITY IMPLEMENTATION

9.1 Authentication Security

Security mechanisms:

- JWT access tokens
 - refresh token support
 - OTP email verification
 - secure logout
-

9.2 Password Security

Implemented using:

- bcrypt hashing
 - salted passwords
 - secure storage
-

9.3 Authorization Security

RBAC middleware validates:

- token authenticity
 - user identity
 - role permissions
-

9.4 API Security

Mechanisms:

- protected routes
 - validation middleware
 - secure response handling
-

9.5 HTTP Security

Libraries:

- Helmet
 - CORS configuration
-

9.6 Abuse Prevention

Protection:

- rate limiting
 - duplicate prevention
 - bot mitigation strategies
-

TESTING

10.1 Testing Objectives

Ensure:

- functional correctness
 - API reliability
 - secure access
 - data consistency
-

10.2 Functional Testing

Modules tested:

- authentication
 - registration
 - login
 - scraping
 - AI extraction
 - dashboards
 - notifications
-

10.3 Integration Testing

Validated:

- frontend ↔ backend communication
 - API ↔ database interaction
 - AI service integration
-

10.4 Security Testing

Validated:

- invalid token rejection
 - unauthorized access prevention
 - malformed request rejection
 - brute force mitigation
-

10.5 Sample Test Cases

Test Case	Expected Result
Valid login	Success
Invalid login	Rejected
Unauthorized API access	Denied
Duplicate job scraping	Blocked
Invalid AI payload	Validation error

DEPLOYMENT

11.1 Deployment Architecture

Suggested deployment:

Frontend:

- Vercel

Backend:

- Render / Railway / AWS

Database:

- MongoDB Atlas

Queue:

- Redis Cloud
-

11.2 Environment Variables

PORT=
MONGO_URI=
JWT_SECRET=
JWT_REFRESH_SECRET=
BREVO_API_KEY=
GEMINI_API_KEY=
GROQ_API_KEY=
REDIS_URL=
NODE_ENV=

RESULTS

Expected Outcomes

System successfully provides:

- hiring intelligence automation
 - AI requirement extraction
 - dashboard analytics
 - recruiter discovery
 - lead scoring
 - secure access
-

Dashboard Outputs

Expected visuals:

- hiring trends
 - skill demand charts
 - company rankings
 - recruiter insights
 - AI extraction summaries
-

FUTURE ENHANCEMENTS

Future extensions:

- chatbot assistant
- predictive hiring intelligence
- advanced CRM integration
- WhatsApp outreach
- automated follow-up sequencing
- resume-job match engine
- ML-based recommendation engine
- multilingual AI parsing
- real-time hiring alerts

CONCLUSION

The **AI Agentic Hiring Intelligence & Recruitment Automation System** presents a scalable, intelligent, and automation-driven solution for modern recruitment intelligence challenges.

The platform successfully combines:

- web scraping
- AI-powered extraction
- analytics
- lead scoring
- recruiter intelligence
- secure dashboard architecture

The system reduces manual effort, improves hiring visibility, enhances operational efficiency, and demonstrates practical implementation of enterprise software engineering concepts.

This project also provides strong exposure to:

- full-stack development
- applied artificial intelligence
- browser automation
- API engineering
- NoSQL schema design
- authentication security
- analytics systems

The architecture remains extensible for future enterprise adoption.

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